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Student Number:.....

2024 Higher School Certificate Trial Examination Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- NESA-approved calculators may be used
- Write your student number on the paper
- Three data sheets and a Periodic Table are provided at the back of this paper

Total marks: 100

Section I – 20 marks

- Attempt Questions 1-20 (pages 2-10)
- Allow about 35 minutes for this section

Section II – 80 marks

- Attempt Questions 21-31 (pages 11-33)
- Allow about 2 hours and 25 minutes for this section

Disclaimer

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Section I – 20 marks

Attempt Questions 1-20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet provided for Questions 1-20

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ A. 2 B. 6 C. 8 D. 9

A ☐ B ☒ C ☐ D ☐

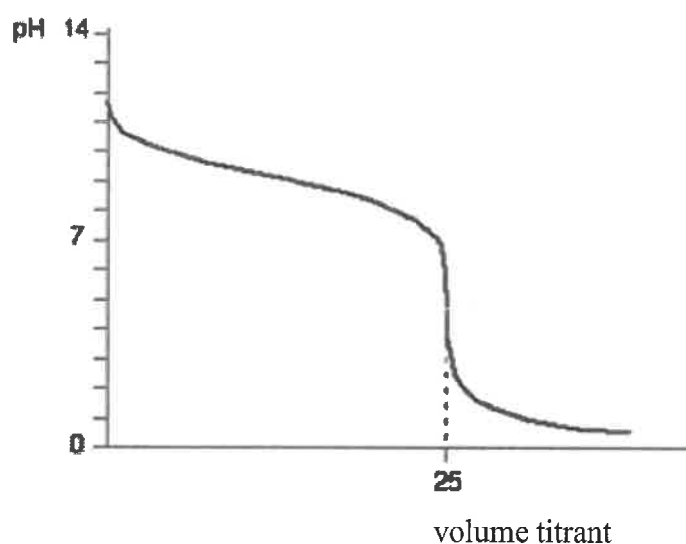
If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you have changed your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ ^{*correct*} C ☐ D ☐

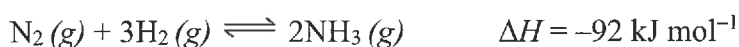
- 1** The diagram below represents a particular acid-base titration.



Which one of the following statements about this titration is correct?

- A. Phenolphthalein is a suitable indicator for this reaction.
- B. A solution of a strong base has been delivered by the burette.
- C. The titrant is a solution of a weak acid.
- D. At the equivalence point, the pH of the solution is close to 4.5.

- 2 What is the systematic name of $\text{CH}_3\text{C}(\text{CH}_3)\text{ClCH}_2\text{CH}_3$?
- A. 1-chloro-1,1-dimethylpropane
 - B. 3-chloro-3-methylbutane
 - C. 2-chloro-2-methylbutane
 - D. 3-chloropentane
- 3 Sodium hydroxide solutions carefully prepared from analytical-grade NaOH pellets cannot be used as primary standards in acid-base titrations. This is because
- A. the NaOH solutions prepared are colourless.
 - B. strong bases are caustic and can cause chemical burns.
 - C. the NaOH pellets are too difficult to dissolve when placed into volumetric flasks.
 - D. the exact concentration of the NaOH solutions made from the pellets is unknown.
- 4 Ammonia can be synthesised by reacting nitrogen and hydrogen according to the following equation:



The rate at which equilibrium is established can be lowered by increasing the

- A. temperature of the reaction mixture at a constant volume.
- B. total volume of the reaction vessel at a constant temperature.
- C. concentration of the reactants at a constant temperature.
- D. pressure in the reaction vessel at a constant temperature.

- 5 For the reaction pathway shown below, which of the following correctly identifies Reagent X and Product Z?



	Reagent X	Product Z
A.	NaOH	1-propyl ethanoate
B.	$\text{H}^+/\text{Cr}_2\text{O}_7^{2-}$	ethyl propanoate
C.	$\text{H}^+/\text{MnO}_4^-$	1-propyl ethanoate
D.	O_2	ethyl ethanoate

- 6 Which one of the following statements about propene is not correct?
- A. It undergoes an addition reaction with hydrogen to form propane.
 - B. It undergoes a polymerisation reaction, expelling water in the process.
 - C. Weak dispersion forces act between propene molecules and consequently it is a gas at room temperature.
 - D. One propene molecule will react with excess oxygen to produce three molecules of water and three molecules of carbon dioxide.
- 7 15.0 mL of 10.0 mol L⁻¹ HCl is added to 60.0 mL of deionised water. What is the concentration of the diluted acid?
- A. 3.33 mol L⁻¹
 - B. 2.50 mol L⁻¹
 - C. 2.00 mol L⁻¹
 - D. 0.500 mol L⁻¹

- 8 A sealed gas syringe contains an equilibrium mixture of brown nitrogen dioxide gas, NO_2 , and colourless dinitrogen tetroxide gas, N_2O_4 .

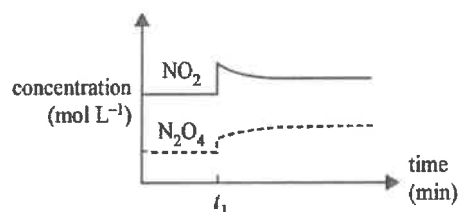
The equation for this equilibrium reaction is:



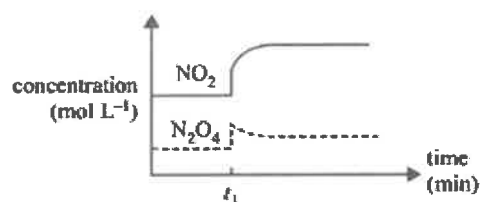
The plunger is suddenly pushed in at time t_1 . The temperature of the gas mixture is kept constant.

Which one of the following concentration–time graphs represents the change that occurs?

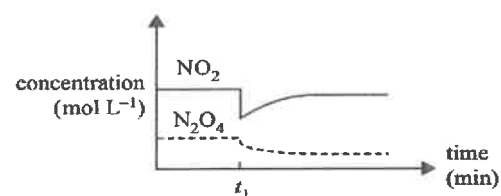
A.



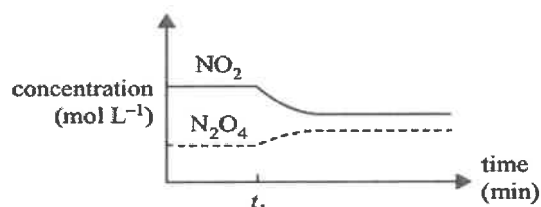
B.



C.



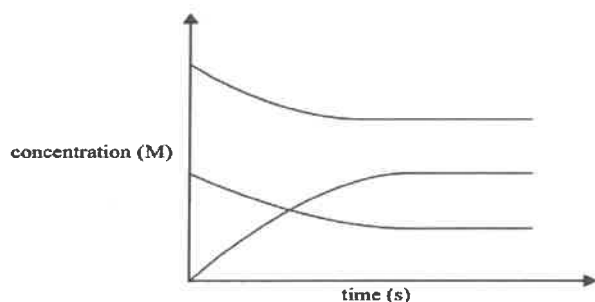
D.



- 9 A condensation reaction involving 200 glucose molecules, $\text{C}_6\text{H}_{12}\text{O}_6$, results in a polysaccharide. What is the molar mass, in g mol^{-1} , of the polysaccharide?
- A. 36000
B. 35982
C. 32418
D. 32400
- 10 The ammonium ion NH_4^+ acts as a weak acid according to the equation:
$$\text{NH}_4^+ (\text{aq}) + \text{H}_2\text{O} (\text{l}) \rightleftharpoons \text{NH}_3 (\text{aq}) + \text{H}_3\text{O}^+ (\text{aq}) \quad K_a = 5.6 \times 10^{-10}$$

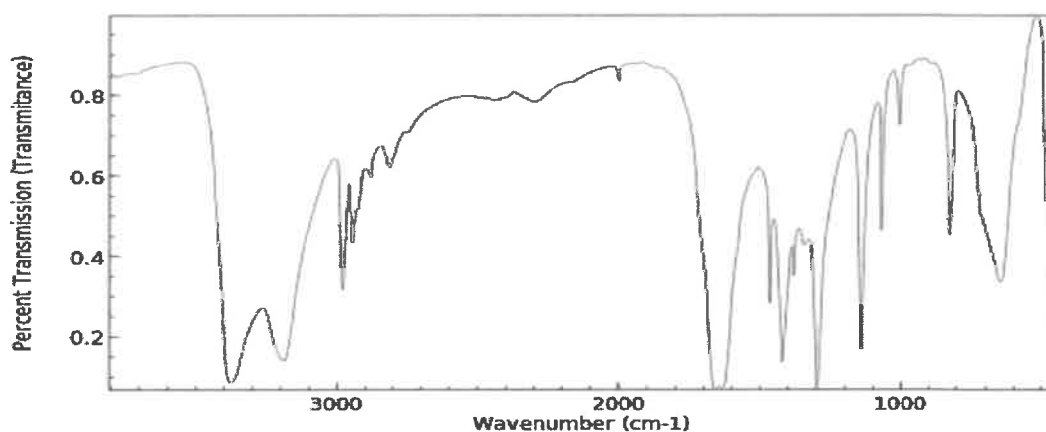
The $[\text{H}_3\text{O}^+]$ of a 0.200 mol L^{-1} ammonium chloride solution is closest to:
- A. $4.79 \times 10^{-6} \text{ mol L}^{-1}$
B. $9.55 \times 10^{-6} \text{ mol L}^{-1}$
C. $1.06 \times 10^{-5} \text{ mol L}^{-1}$
D. $1.51 \times 10^{-5} \text{ mol L}^{-1}$
- 11 What is the name of the product formed when chlorine, Cl_2 , reacts with but-1-ene?
- A. 1,2-dichlorobutane
B. 1,4-dichlorobutane
C. 2,2-dichlorobutane
D. 2,3-dichlorobutane

- 12 Consider the following concentration versus time graph:



Which one of the following reactions could produce the graph shown above?

- A. $\text{N}_2(g) + 3\text{H}_2(g) \rightleftharpoons 2\text{NH}_3(g)$
- B. $\text{H}_2(g) + \text{Br}_2(g) \rightleftharpoons 2\text{HBr}(g)$
- C. $2\text{HCl}(g) \rightleftharpoons \text{H}_2(g) + \text{Cl}_2(g)$
- D. $2\text{SO}_3(g) \rightleftharpoons 2\text{SO}_2(g) + \text{O}_2(g)$
- 13 An organic compound Z does not react as an acid or a base. The solid dissolves readily in water at 25°C, forming strong hydrogen bonds. The infra-red spectrum of compound Z is shown below:



NIST Chemistry WebBook (<https://webbook.nist.gov/chemistry>)

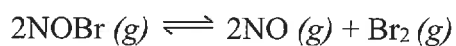
The compound could be

- A. $\text{CH}_3\text{CH}_2\text{COOH}$
- B. $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
- C. $\text{CH}_3\text{CH}_2\text{CONH}_2$
- D. $\text{NH}_2\text{CH}_2\text{COOH}$

14 Which molecule can exist as a *cis* and *trans* isomer?

- A. Hex-1-ene
- B. Hex-2-ene
- C. 2-methylpent-1-ene
- D. 2-methylbut-2-ene

15 Nitrosyl bromide decomposes according to the following equation:



A 0.32 mol sample of NOBr is placed in an evacuated 1.00 L flask. After the system comes to equilibrium, the flask contains 0.23 mol NOBr.

What are the concentrations of NO and Br₂ in the flask at equilibrium?

	[NO] mol L ⁻¹	[Br ₂] mol L ⁻¹
A.	0.09	0.045
B.	0.09	0.09
C.	0.18	0.09
D.	0.46	0.23

- 16 A 2.0 g sample of silver chloride ($MM = 143.4 \text{ g mol}^{-1}$) was added to 100.0 mL of water in a beaker.

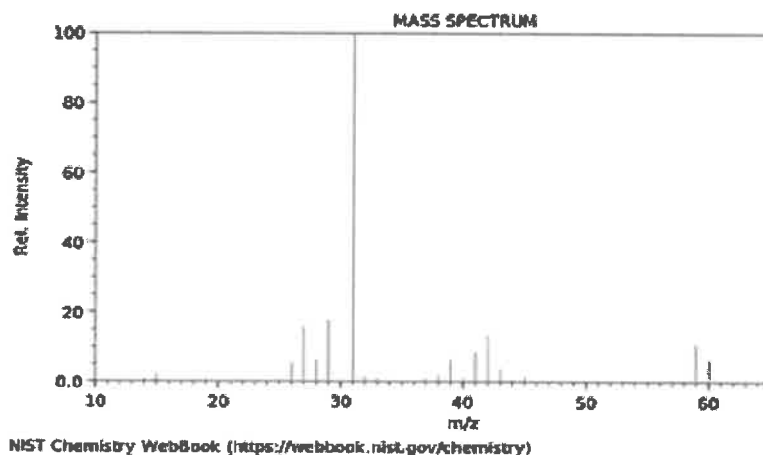
The solubility of silver chloride at this temperature is $1.77 \times 10^{-10} \text{ mol L}^{-1}$.

The sample was then diluted by adding another 100.0 mL of water.

What is the ratio of the concentration of silver ions in solution before and after dilution?

- A. 1:1
 - B. 1:2
 - C. 2:1
 - D. 4:1
- 17 Which of the following is an Arrhenius base?
- A. Na
 - B. Ba(OH)_2
 - C. Na_2CO_3
 - D. HPO_4^{2-}
- 18 Which pair of ions can be distinguished by addition of chloride ions?
- A. Na^+ and K^+
 - B. Ba^{2+} and Ca^{2+}
 - C. Mg^{2+} and Ca^{2+}
 - D. Pb^{2+} and Zn^{2+}

- 19 An unidentified organic substance is found to react with an acid to form an ester. The mass spectrum of this compound is shown below.



Which one of the following species is consistent with the base peak on the mass spectrum?

- A. $[\text{CH}_2\text{OH}]^+$
 - B. $[\text{CH}_3\text{CH}_2\text{O}]^+$
 - C. $[\text{CH}_3\text{O}]^+$
 - D. $[\text{COOH}]^+$
- 20 The ^{13}C NMR spectrum of an organic compound has two peaks. The ^1H NMR spectrum of the same compound has only 1 peak, with no splitting. The compound is
- A. propanal.
 - B. propanoic acid.
 - C. propane.
 - D. propanone.

Student Number:

Chemistry

Section II – 80 marks

Attempt Questions 21-31

Allow about 2 hours and 25 minutes for this section

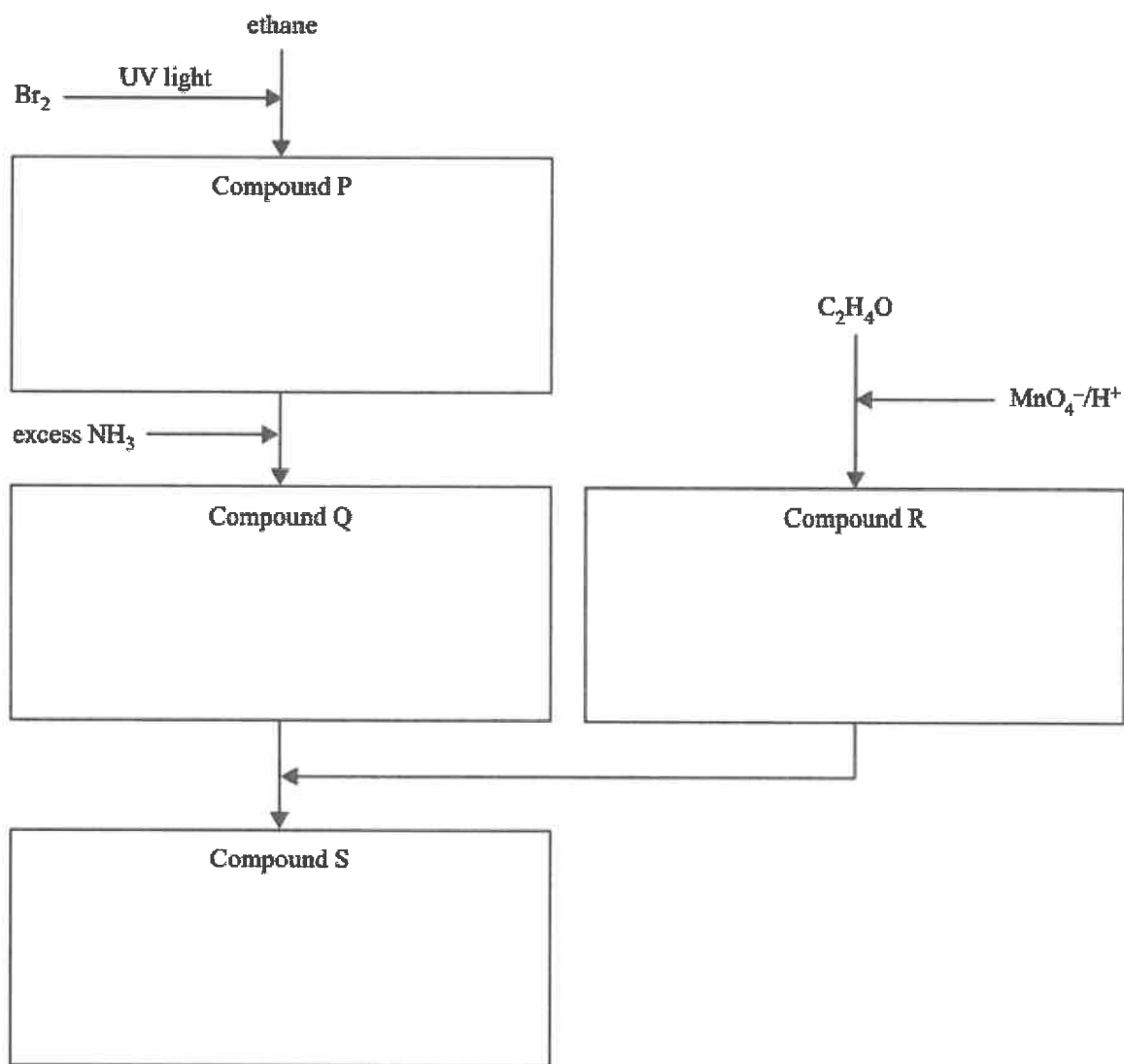
Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Extra writing space is provided on pages 34 and 35. If you use this space, clearly indicate which question you are answering.

Question 21 (8 marks)

The diagram below shows a reaction pathway involving carbon compounds, starting with ethane and $\text{C}_2\text{H}_4\text{O}$. Assume that the ethane and bromine react in a 1:1 mole ratio and that only 1 compound containing carbon atoms is formed.



- (a) Write the IUPAC name of Compound P in the box provided. 1
- (b) Name the type of reaction that produces Compound Q from Compound P. 1
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- (c) Write the structural formula of Compound Q in the box provided. 1

Question 21 continues on the next page

Question 21 (continued)

- (d) When $\text{C}_2\text{H}_4\text{O}$ is mixed with potassium permanganate, KMnO_4 , and sulfuric acid, H_2SO_4 , a carboxylic acid, Compound R, is produced. 2

Draw the structure of Compound R in the box provided and name Compound R below.

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- (e) What is the function of the H_2SO_4 in the reaction to produce Compound R? 1

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- (f) Compound Q reacts with Compound R in a condensation reaction to produce Compound S. 2

Draw the structural formula for Compound S in the box provided and name Compound S below.

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Question 22 (4 marks)

A scientist used quantitative analysis of an unknown compound **A**, which contains only carbon, oxygen, and chlorine, to determine its molecular formula.

- (a) A 3.00 g sample of **A** was completely vaporised in a 1.00 L container at 70.0°C and exerted a pressure of 86.51 kPa. What is the molar mass of **A**? 1

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- (b) A 0.300 g sample of **A** was added to 100 mL water, which resulted in the conversion of all the chlorine in **A** to HCl. After briefly bubbling nitrogen gas through the solution, the hydrochloric acid was titrated with 0.2000 mol L⁻¹ NaOH solution. 2

The titration required 30.33 mL of added NaOH to reach a phenolphthalein endpoint. What is the mass percentage of chlorine in compound **A**?

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- (c) What is the molecular formula for **A**? 1

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Question 23 (9 marks)

The following table lists the pH of 0.10 mol L^{-1} solutions of four different acids at 25°C .

Acid	pH
I	1.0
II	3.0
III	0.7
IV	2.1

- (a) Which one of the four acids listed in the table above has the smallest K_a value? 1

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- (b) Which acid must have more than one acidic hydrogen per molecule? Give a reason for your answer. 1

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- (c) Using the concentration and the pH of acid IV, calculate the percentage ionisation of acid IV in the 0.10 mol L^{-1} solution. 1

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Question 23 continues on the next page

Question 23 (continued)

- (d) Samples of the solutions of acids I and IV are diluted by a factor of 10. For which of these acids is the change in pH greater? Explain your answer.

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- (e) Methanoic acid is a weak monoprotic acid, with $K_a = 1.8 \times 10^{-4}$. Calculate the concentration of a methanoic acid solution that will have the same pH as acid IV.

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- (f) The dissociation of methanoic acid in water is exothermic. If a solution of the acid is heated, will the pH of the solution increase, decrease or remain constant? Give an explanation for your answer.

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Question 24 (6 marks)

Polymers are significant organic molecules. They are classified on the basis of the method of linking of their monomers. Choose TWO polymers you have studied which have different methods of linkage.

For each of these polymers:

- Name the polymer
- Classify the polymer according to its type of linkage
- Draw a monomer or monomers
- Draw part of a polymer chain that includes at least 4 monomer units

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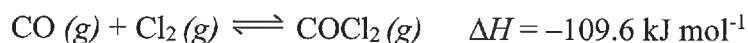
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Question 25 (8 marks)

Phosgene, COCl_2 , is a highly toxic gas used in chemical industry. COCl_2 is usually produced by the reaction between carbon monoxide and chlorine according to the equation.



A scientist conducted two experiments on the equilibrium reaction of COCl_2 .

The experiments were conducted in evacuated and sealed 4 L containers at 150°C .

Experiment 1: 2 mol of $\text{COCl}_2(\text{g})$ was injected into a previously evacuated, sealed 4 L container.

Experiment 2: 4 mol of $\text{COCl}_2(\text{g})$ was injected into another previously evacuated, sealed 4 L container.

- (a) Which experiment had the higher initial rate of production of Cl_2 ? Justify your answer. 2

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- (b) If, for Experiment 1, the concentrations of $\text{COCl}_2(g)$ and $\text{CO}(g)$ were equal at equilibrium, $[\text{COCl}_2] = [\text{CO}]$, then what conclusion could be made about the relative concentrations of COCl_2 and CO in Experiment 2 at equilibrium? Justify your answer. 3

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Question 25 continues on the next page

Question 25 (continued)

- (c) 2 mol of an inert gas is injected into the container in Experiment 1. The temperature is kept at 150°C. What effect will this have on the rate of production of the Cl_2 in the container? Justify your answer using collision theory.

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- (d) The temperature for Experiment 2 is increased to 200°C. Explain the effect on the equilibrium concentration of COCl_2 in the reaction.

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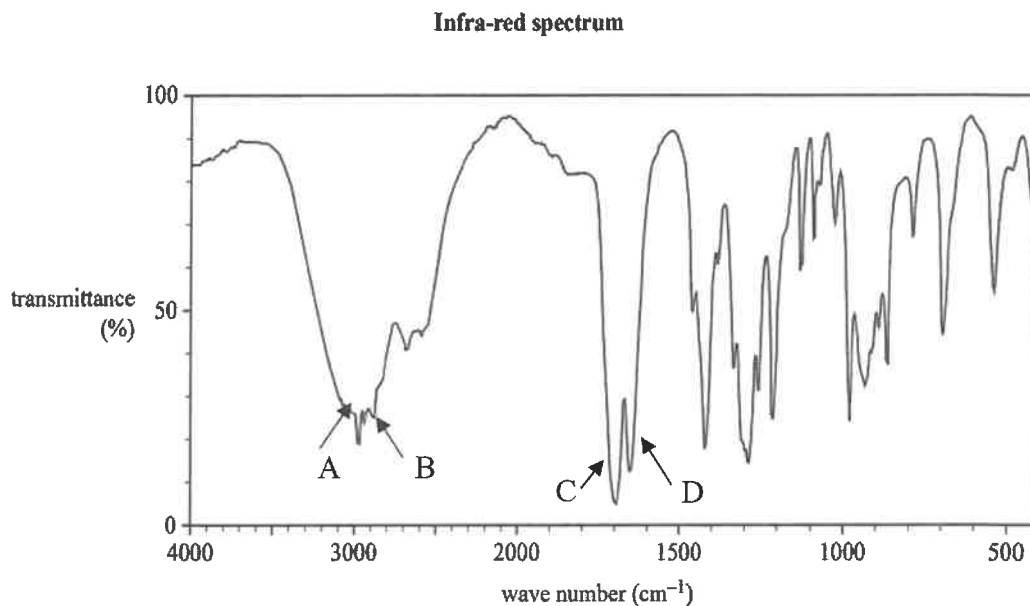
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Question 26 (13 marks)

A student investigated an organic substance, Compound Y, with the molecular formula $C_5H_8O_2$.

- (a) The infra-red spectrum of Compound Y is shown below. On the spectrum, FOUR peaks have been labelled.

3



Data: SDBSWeb, <<http://sdbs.db.aist.go.jp>>,
National Institute of Advanced Industrial Science and Technology

Identify the bonds responsible for each of the 4 peaks identified on the spectrum.

A =

B =

C =

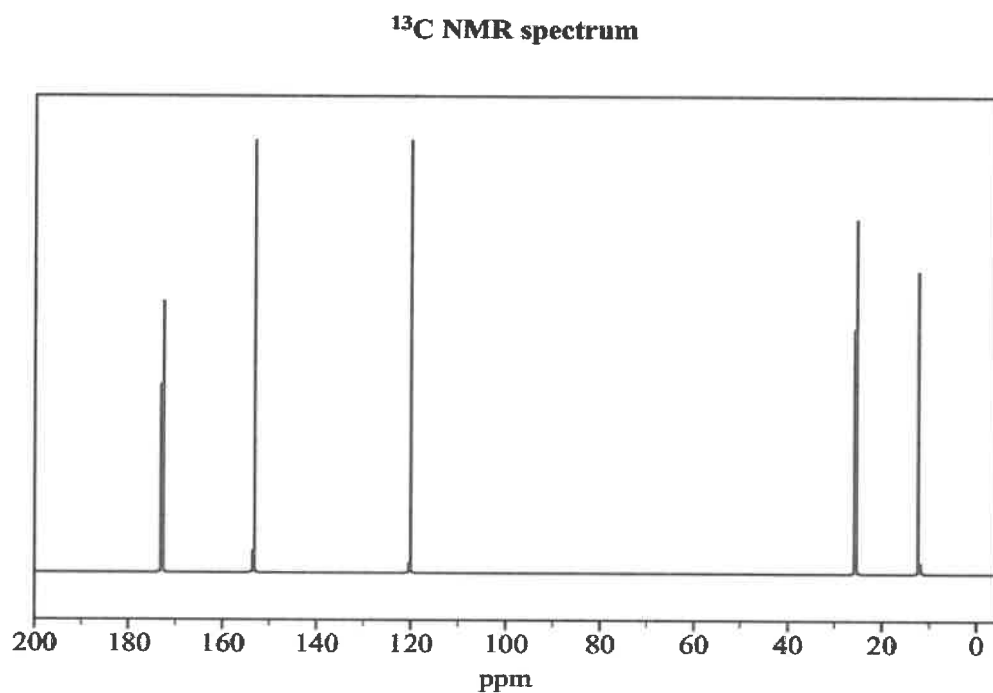
D =

Question 26 continues on the next page

Question 26 (continued)

- (b) A sample of Compound Y was further analysed using ^{13}C NMR spectroscopy. The spectrum is shown below.

2



Data: SDBSWeb, <<http://sdb.s.db.aist.go.jp>>,
National Institute of Advanced Industrial Science and Technology

Use the information provided in the ^{13}C NMR spectrum to identify the number of different carbon environments for Compound Y and explain the significance of the difference in the position of the peaks on the spectrum.

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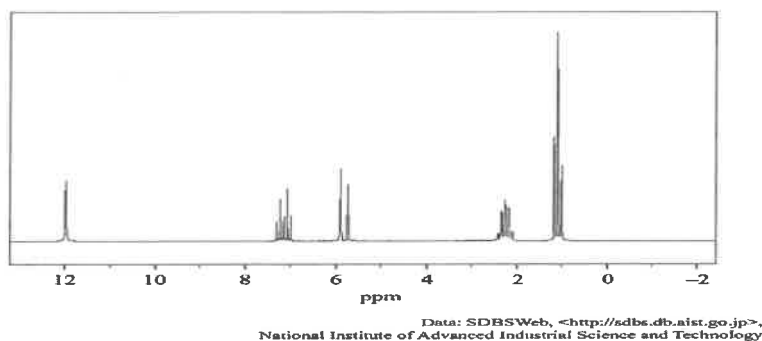
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Question 26 continues on the next page

Question 26 (continued)

- (c) A sample of Compound Y was further analysed using ^1H NMR spectroscopy. The spectrum is shown below.

2



^1H NMR data

Chemical shift (ppm)	Relative peak area	Peak splitting
1.1	3	triplet (3)
2.3	2	pentet (5)
5.8	1	doublet (2)
7.1	1	quartet (4)
12	1	singlet (1)

For the signal at 1.1 ppm in the ^1H NMR spectrum, identify what specific information is provided by the relative peak area and the peak splitting.

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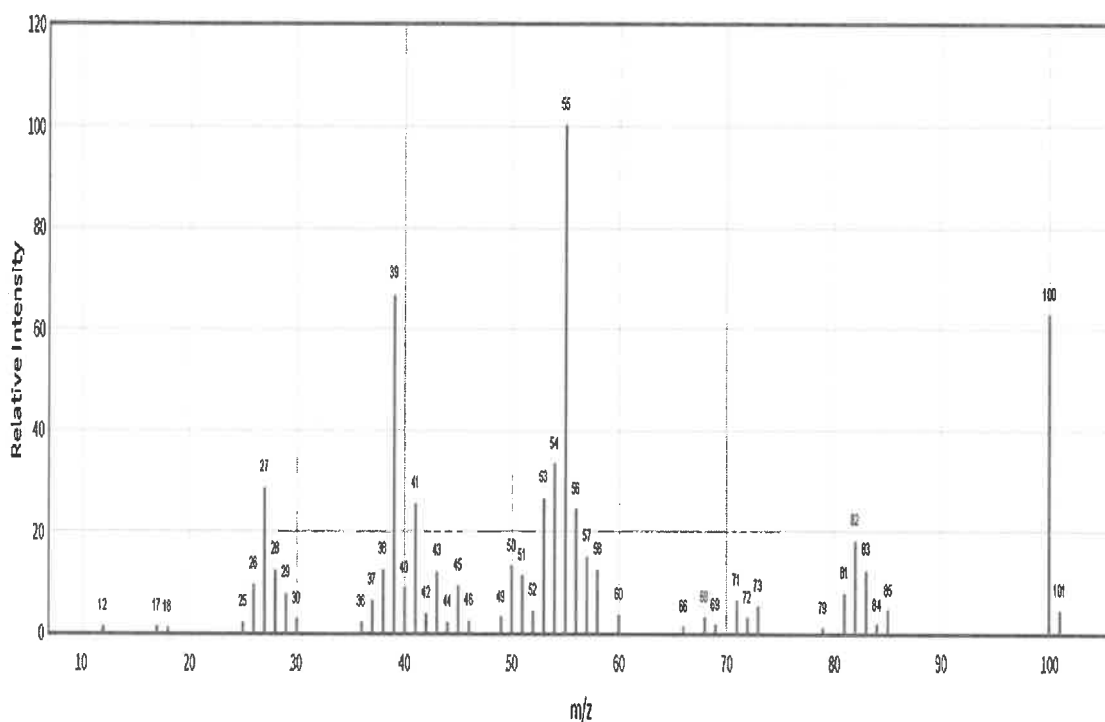
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Question 26 continues on the next page

Question 26 (continued)

(d) The mass spectrum of Compound Y is shown below.

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Identify the parent peak and the base peak and suggest the structure of the fragment formed which has the m/z value shown at the base peak.

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Question 26 continues on the next page

Question 26 (continued)

- (e) Draw the structural formula of Compound Y. Justify your answer by relating aspects of the structure to the evidence from the spectra and data table. 4

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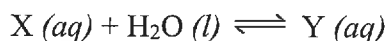
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Question 27 (6 marks)

When substance X is dissolved in water, it reacts to form an equilibrium mixture with substance Y according to the equation:



The concentration of X can be determined using UV-visible spectroscopy. X absorbs strongly at 290 nm. Y shows no absorption at this wavelength.

In a particular experimental arrangement at 25°C, the relationship between absorbance at 290 nm and concentration of X in mol L⁻¹ is given by:

$$\text{Absorbance} = 3.75 \times [\text{X}]$$

In the experiment, 0.110 mol of X is dissolved rapidly in 1.00 L of water at 25°C. The absorbance of the solution changes as some of the X is converted to Y. The table below shows the change in absorbance over time (measured in seconds).

Absorbance	0.389	0.274	0.244	0.230	0.226	0.226
Time (s)	6.00	60.0	90.0	120	240	480

- (a) Calculate the concentration of X, in mol L⁻¹, when the reaction reached equilibrium. 1

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- (b) Calculate the absorbance at the instant that X was dissolved in the water before any reaction occurred. 1

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- (c) Calculate the percentage of the original 0.110 mol of X that has been converted into Y at equilibrium. 2

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Question 27 continues on the next page

Question 27 (continued)

- (d) The average rate of a reaction can be determined by calculating the change in concentration of a reactant per second. Calculate the average rate, in $\text{mol L}^{-1} \text{s}^{-1}$, at which the concentration of X changed during the first 6.00 s of the reaction. 2

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Question 28 (5 marks)

- (a) Calculate the mass of lead(II) fluoride ($K_{sp} = 4.0 \times 10^{-8}$) which will dissolve in 1.00 L of water. 2

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- (b) Determine the molar solubility of lead(II) fluoride in 1.00 L of solution to which 0.10 mol of solid lead(II) nitrate has been added. 2

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- (c) The solubility of lead(II) fluoride increases with temperature. 1

Write an equation for the equilibrium reaction of lead(II) fluoride in water and include information about the enthalpy change for this reaction.

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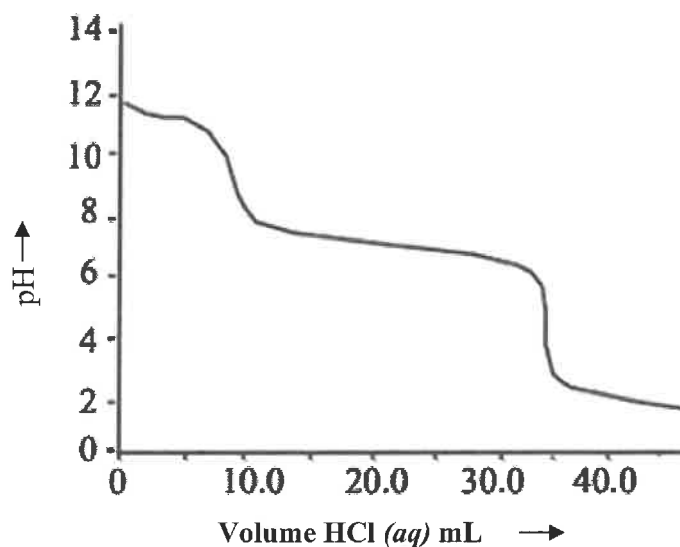
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Question 29 (10 marks)

The percentages of NaHCO_3 and Na_2CO_3 are to be determined in a mixture of them with KCl.

A 0.500 g sample of the mixture is dissolved in 50.0 mL of deionised water and titrated with $0.115 \text{ mol L}^{-1} \text{ HCl}$. A student produced this sketch of the pH titration curve.



- (a) Write balanced equations for the reaction that is responsible for the equivalence point that occurs at about $\text{pH} = 9$ and $\text{pH} = 5$. 2

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- (b) Calculate the total number of moles of acid used to reach each equivalence point if the volumes are 9.63 mL and 34.27 mL, respectively. 2

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Question 29 continues on the next page

Question 29 (continued)

- (c) Determine the number of grams of Na_2CO_3 , NaHCO_3 and KCl and their percentages in the original mixture. 4

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- (d) Sketch a titration curve for a solution of Na_2CO_3 by itself and describe how it differs from the given curve. 2

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Question 30 (6 marks)

Consider the reaction shown in the following equation:



- (a) Write an expression for the equilibrium constant for this reaction.

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- (b) 10.0 mol of NOBr, 10.0 mol of NO and 5.0 mol of Br₂ are placed in a 1.0 L container at 1000 K. Predict in which direction the reaction will proceed. Justify your answer.

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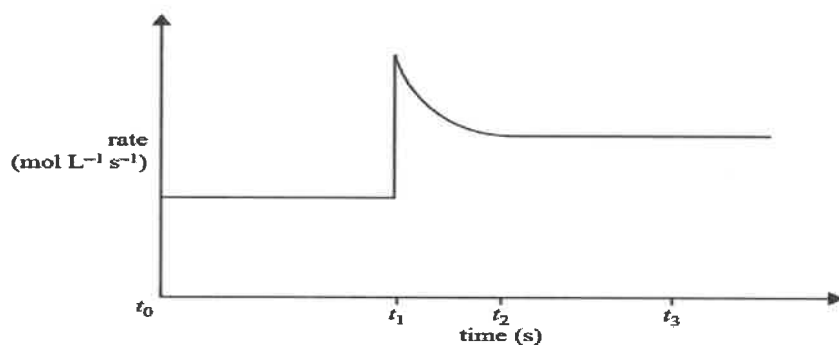
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Question 30 continues on the next page

Question 30 (continued)

- (c) A mixture of NO, NOBr and Br₂ is initially at equilibrium. The following graph shows how the rate of formation of NOBr in the mixture changes when the volume of the reaction vessel is decreased at time t_1 . 3



Use collision theory and factors that affect the rate of a reaction to explain the shape of the graph at the time intervals indicated.

- (i) between t_0 and t_1

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- (ii) at t_1

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- (iii) between t_1 and t_2

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Question 31 (5 marks)

- (a) Write an equation for the complete combustion of octane, C_8H_{18} . 1

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- (b) Write an equation for the fermentation of glucose to produce ethanol. 1

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- (c) What are “biofuels”? Outline 2 benefits and 2 problems associated with the use of a named biofuel. 3

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End of Examination

Section II - Extra writing space

If you use this space, clearly indicate which question you are answering.

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Section II - Extra writing space

If you use this space, clearly indicate which question you are answering.

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Student Number:

CHEMISTRY – MULTIPLE-CHOICE ANSWER SHEET

ATTEMPT ALL QUESTIONS

Question	1	A ☐	B ☐	C ☐	D ☐
	2	A ☐	B ☐	C ☐	D ☐
	3	A ☐	B ☐	C ☐	D ☐
	4	A ☐	B ☐	C ☐	D ☐
	5	A ☐	B ☐	C ☐	D ☐
	6	A ☐	B ☐	C ☐	D ☐
	7	A ☐	B ☐	C ☐	D ☐
	8	A ☐	B ☐	C ☐	D ☐
	9	A ☐	B ☐	C ☐	D ☐
	10	A ☐	B ☐	C ☐	D ☐
	11	A ☐	B ☐	C ☐	D ☐
	12	A ☐	B ☐	C ☐	D ☐
	13	A ☐	B ☐	C ☐	D ☐
	14	A ☐	B ☐	C ☐	D ☐
	15	A ☐	B ☐	C ☐	D ☐
	16	A ☐	B ☐	C ☐	D ☐
	17	A ☐	B ☐	C ☐	D ☐
	18	A ☐	B ☐	C ☐	D ☐
	19	A ☐	B ☐	C ☐	D ☐
	20	A ☐	B ☐	C ☐	D ☐

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Chemistry

FORMULAE SHEET

$$n = \frac{m}{MM}$$

$$q = mc\Delta T$$

$$pK_a = -\log_{10}[K_a]$$

$$c = \frac{n}{V}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$A = \epsilon lc = \log_{10} \frac{I_0}{I}$$

$$PV = nRT$$

$$\text{pH} = -\log_{10}[\text{H}^+]$$

Avogadro constant, N_A $6.022 \times 10^{23} \text{ mol}^{-1}$

Volume of 1 mole ideal gas: at 100 kPa and

at 0°C (273.15 K) 22.71 L

at 25°C (298.15 K) 24.79 L

Gas constant $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

Ionisation constant for water at 25°C (298.15 K), K_w 1.0×10^{-14}

Specific heat capacity of water $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

DATA SHEET

Solubility constants at 25°C

Compound	K_{sp}	Compound	K_{sp}
Barium carbonate	2.58×10^{-9}	Lead(II) bromide	6.60×10^{-6}
Barium hydroxide	2.55×10^{-4}	Lead(II) chloride	1.70×10^{-5}
Barium phosphate	1.3×10^{-29}	Lead(II) iodide	9.8×10^{-9}
Barium sulfate	1.08×10^{-10}	Lead(II) carbonate	7.40×10^{-14}
Calcium carbonate	3.36×10^{-9}	Lead(II) hydroxide	1.43×10^{-15}
Calcium hydroxide	5.02×10^{-6}	Lead(II) phosphate	8.0×10^{-43}
Calcium phosphate	2.07×10^{-29}	Lead(II) sulfate	2.53×10^{-8}
Calcium sulfate	4.93×10^{-5}	Magnesium carbonate	6.82×10^{-6}
Copper(II) carbonate	1.4×10^{-10}	Magnesium hydroxide	5.61×10^{-12}
Copper(II) hydroxide	2.2×10^{-20}	Magnesium phosphate	1.04×10^{-24}
Copper(II) phosphate	1.40×10^{-37}	Silver bromide	5.35×10^{-13}
Iron(II) carbonate	3.13×10^{-11}	Silver chloride	1.77×10^{-10}
Iron(II) hydroxide	4.87×10^{-17}	Silver carbonate	8.46×10^{-12}
Iron(III) hydroxide	2.79×10^{-39}	Silver hydroxide	2.0×10^{-8}
Iron(III) phosphate	9.91×10^{-16}	Silver iodide	8.52×10^{-17}
		Silver phosphate	8.89×10^{-17}
		Silver sulfate	1.20×10^{-5}

Infrared absorption data

Bond	Wavenumber/cm ⁻¹
N—H (amines)	3300–3500
O—H (alcohols)	3230–3550 (broad)
C—H	2850–3300
O—H (acids)	2500–3000 (very broad)
C≡N	2220–2260
C=O	1680–1750
C=C	1620–1680
C—O	1000–1300
C—C	750–1100

¹³C NMR chemical shift data

Type of carbon	δ/ppm
$\begin{array}{c} \quad \\ -C-C- \\ \quad \end{array}$	5–40
$\begin{array}{c} \\ R-C-Cl \text{ or } Br \\ \end{array}$	10–70
$\begin{array}{c} \\ R-C-C- \\ \quad \\ O \end{array}$	20–50
$\begin{array}{c} \\ R-C-N \\ \quad \diagup \quad \diagdown \end{array}$	25–60
$\begin{array}{c} \\ -C-O- \\ \end{array}$ alcohols, ethers or esters	50–90
$\begin{array}{c} \diagup \quad \diagdown \\ C=C \\ \diagdown \quad \diagup \end{array}$	90–150
R—C≡N	110–125
	110–160
$\begin{array}{c} R-C- \\ \\ O \end{array}$ esters or acids	160–185
$\begin{array}{c} R-C- \\ \\ O \end{array}$ aldehydes or ketones	190–220

UV absorption

(This is not a definitive list and is approximate.)

Chromophore	λ _{max} (nm)
C—H	122
C—C	135
C=C	162

Chromophore	λ _{max} (nm)
C≡C	173 178 196 222
C—Cl	173
C—Br	208

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

1	H	1.008	Hydrogen
3	Li	6.941	Lithium
11	Na	22.99	Sodium
19	K	39.10	Potassium
37	Rb	85.47	Rubidium
55	Cs	132.9	Caesium
87	Fr		Francium

Atomic Number	79
Symbol	Au
Standard Atomic Weight	197.0
Name	Gold

5	6	7	8	9	10
B	C	N	O	F	Ne
10.81	12.01	14.01	16.00	19.00	20.18
Boron	Carbon	Nitrogen	Oxygen	Fluorine	Neon
13	14	15	16	17	18
Al	Si	P	S	Cl	Ar
26.98	28.09	30.97	32.07	35.45	39.95
Aluminum	Silicon	Phosphorus	Sulfur	Chlorine	Argon
31	32	33	34	35	36
Ga	Ge	As	Se	Br	Kr
69.72	72.64	74.92	78.96	79.90	83.80
Gallium	Germanium	Arsenic	Selenium	Bromine	Krypton
49	50	51	52	53	54
In	Sn	Sb	Te	I	Xe
114.8	118.7	121.8	127.6	126.9	131.3
Indium	Tin	Antimony	Tellurium	Iodine	Xenon
81	82	83	84	85	86
Tl	Pb	Bi	Po	At	Rn
204.4	207.2	209.0			
Thallium	Lead	Bismuth	Polonium	Astatine	Radon
113	114	115	116	117	118
Nh	Fl	Mc	Lv	Ts	Og
Nihonium	Flerovium	Moscovium	Livermorium	Tennessee	Oganesson

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57	58	59	60	61	62	63	64	65	66	67	68	69	70	71
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
138.9	140.1	140.9	144.2		150.4	152.0	157.3	158.9	162.5	164.9	167.3	168.9	173.1	175.0
Lanthanum	Cerium	Praseodymium	Neodymium	Promethium	Samarium	Europium	Gadolinium	Terbium	Dysprosium	Holmium	Erbium	Thulium	Ytterbium	Lutetium

89	90	91	92	93	94	95	96	97	98	99	100	101	102	103
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
Actinium	Thorium	Protactinium	Uranium	Neptunium	Plutonium	Americium	Curium	Berkelium	Californium	Einsteinium	Fermium	Mendelevium	Nobelium	Lawrencium

Elements with no reported values in the table have no stable nuclides.

The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data and elements (November 2010 version) presented in this study. The data were obtained from the IUPAC Periodic Table of the Elements (November 2010 version) and the IUPAC Periodic Table of the Elements (February 2010 version).